

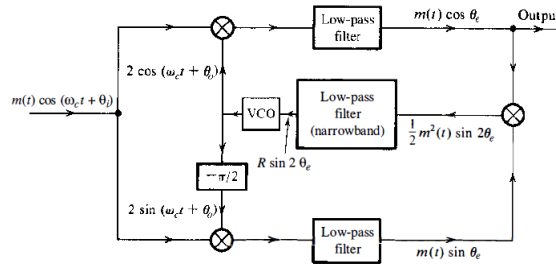
Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2022-2023
Mid-Semester Exam (Close Book)

EEE F311 Communication Systems

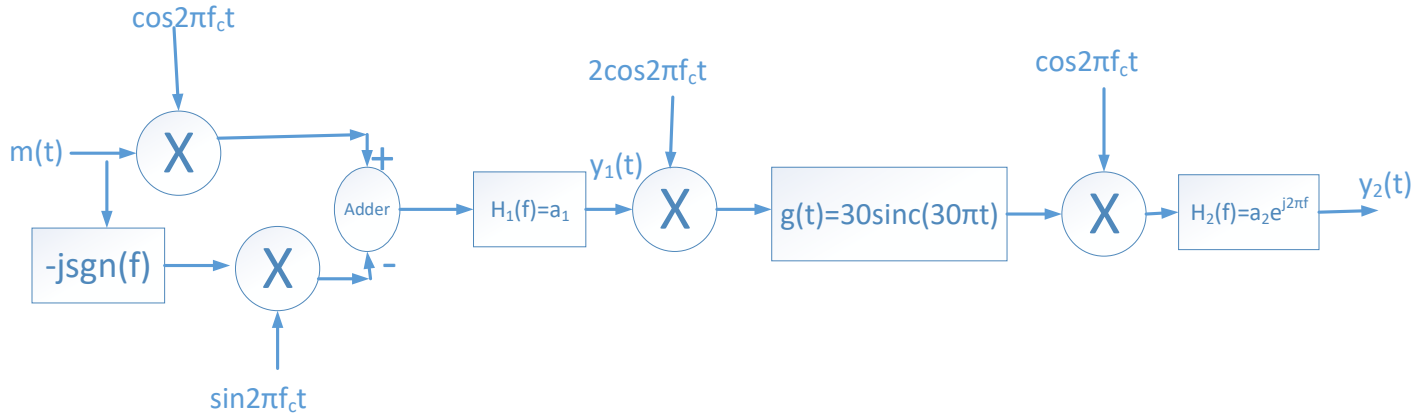
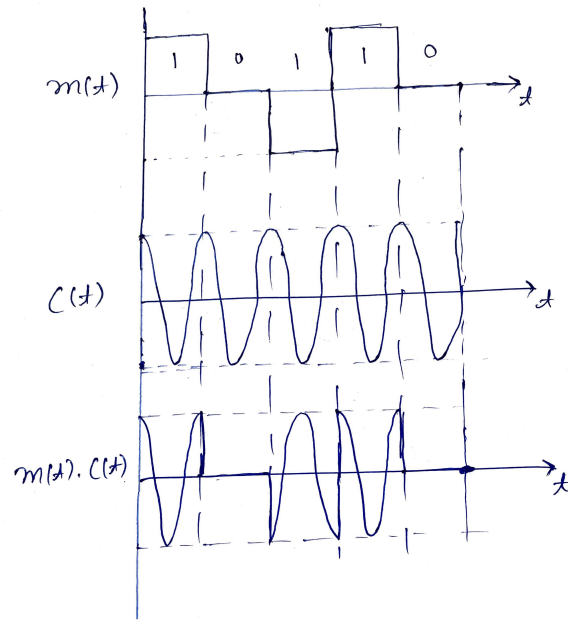
Date: 05-11-2022, Duration: 90 Minutes, Maximum Marks: 50

Make suitable assumptions, if any. Put the final answers in a box.

1. (a) $m(t) = 2 + 4 \sin 20\pi t$ amplitudes modulates a carrier signal $c(t) = 6 \cos 100\pi t$. Find the power efficiency (in %) of the modulated AM signal. [Marks: 2]
Sol: 25%.
- (b) A NBFM signal has a carrier frequency of 100 Hz and frequency deviation of 2 Hz. Find the frequency deviation if the NBFM is passed through a mixer at 50 Hz and multiplier having a frequency doubler. [Marks: 2]
Sol: 4 Hz.
- (c) A broadcasting system has a spectrum in the range of 2000KHz-4000KHz and employs superhetrodyne receiver. Find the minimum IF frequency such that image frequency falls beyond the allocated spectrum. [Marks: 2]
Sol: 1000 Hz.
- (d) Draw the Costas loop for the demodulation of DSB-SC signal. [Marks: 2]
Sol:



- (e) A message bit stream 10110 is line encoded using bipolar NRZ to generate pulse stream $m(t)$. Sketch $m(t)$ considering a rectangular pulse width of $\frac{1}{5}$ second and amplitude 1. Also sketch $m(t) \cos 4\pi t$ for 1 second. [Marks: 2]
Sol:
2. The signal $m(t)$ in the following figure has a bandwidth 10 Hz.
 - (a) Find the signal $y_1(t)$ in terms of the input signal $m(t)$. [Marks: 3]
Sol: $a_1 m(t)$.



- (b) Find the output signal $y_2(t)$ in terms of the input signal $m(t)$. [Marks: 3]

Sol: $a_1 a_2 m(t-1) \cos 2\pi f_c(t-1)$.

- (c) Apply synchronous detector on $y_2(t)$. [Marks: 4]

Sol: multiply $y_2(t)$ by $\frac{2}{a_1 a_2} \cos 2\pi f_c(t-1)$, then a low pass filter of BW 10 Hz to get $m(t-1)$.

3. An FM signal is given by $x(t) = 50 \cos(2000\pi t + 2 \sin 200\pi t)$ with the parameter $k_f = 2$ Hz/Volt. [Marks: 2×5]

- (a) Find the message signal $m(t)$.

Sol: $100 \cos 200\pi t$.

- (b) Find the modulation index β of the FM signal.

Sol: 2.

- (c) Find the bandwidth of the FM signal using Carson's rule.

Sol: 600 Hz.

- (d) An alternate representation of $x(t) = 50 \cos(2000\pi t + 2 \sin 200\pi t)$ is given by $x(t) = \sum_{n=-\infty}^{\infty} 50 J_n(\beta) \cos(2\pi(f_c + n f_m)t)$, where f_c is the carrier frequency and f_m is the message

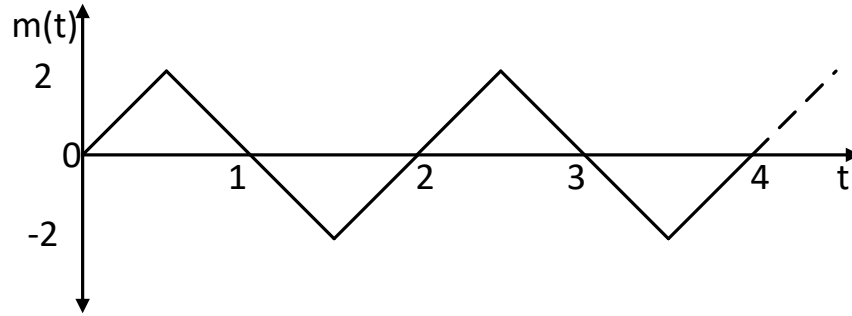
frequency and β is the modulation index. Plot $|X(f)|$ (in terms of Bessel function) for $n = 0$ and $n = 1$ clearly labeling the x-axis and y-axis.

Sol: There will be two impulses of amplitude $25J_0(2)$ at 1000 Hz and -1000 Hz. There will be another two impulses of amplitude $25J_1(2)$ at 1100 Hz and $25J_{-1}(2)$ at -1100 Hz.

- (e) The signal $x(t)$ is passed through a channel $h(t) = 100\text{sinc}(100\pi t) \cos 2000\pi t$. Find the output power in terms of Bessel function.

Sol: $50J_0(2) \cos 2\pi f_c t$, which the power is $1250J_0^2(2)$

4. A periodic message signal $m(t)$ (as shown in figure) is to be converted to binary bit-stream using PCM and DM techniques.



- (a) The signal is sampled at a rate of 1000 samples per second and encoded using the PCM. Find the minimum bandwidth required to transmit the data to achieve an SQNR of 128. [Marks: 5]

Sol: Average Power = $\frac{4}{3}$, $m_p = 2$, $SQNR = 3L^2 \frac{P_m}{m_p^2}$, which gives $L = 16 = 4$ bits/sample. Total data rate = 4000 bits/sec. BW=2000 Hz.

- (b) Find the minimum step-size of the DM to avoid slope-over load condition achieving the same data rate transmission as that of the PCM (as obtained in part (a)). [Marks: 5]

Sol: $f_s = 4000$ samples/sec. $\frac{dm_t}{dt} = 4$, $\Delta = \frac{1}{1000}$.

5. (a) Use the time-domain criteria to show that the pulse $p(t) = \frac{\cos(\pi R_s t)}{1-(2R_s t)^2} \text{sinc}(\pi R_s t)$ satisfies the Nyquist criteria for zero ISI. [Marks: 3]

Sol: Take the limit $t = 0$ to get $p(t) = 1$ and $p(t) = 0$ when $t \neq 0$.

- (b) How much time (in seconds) will be required to transfer a movie data of 1 GB over a bandwidth of 10 MHz using Nyquist pulse $p(t) = \frac{\cos(\pi R_s t)}{1-(2R_s t)^2} \text{sinc}(\pi R_s t)$ with NRZ binary polar encoding, where R_s denotes the transmission rate. Take 1 GB = 10^9 bits. [Marks: 3]

Sol: $R_s = B$, since $\alpha = 1$. Time= 100 seconds.

- (c) A message signal $m(t) = \sum p(t - nT)$, where $p(t) = \frac{\cos(\pi \alpha R_s t)}{1-(2\alpha R_s t)^2} \text{sinc}(\pi R_s t)$ is transmitted with pulse rate satisfying Nyquist criterion for zero-ISI. Sketch the frequency response of the pulse i.e., $P(f)$ (both-sided) for $n = 0$ and $n = 1$ in the same plot satisfying the Nyquist criteria for zero ISI for (i) $\alpha = 0$ and (ii) $\alpha = 1$. [Marks: 2+2]

Sol:

